Starter Retrieval Grid: Genetics & Gene Delivery Recall

Do Now Activity (5 Minutes) · Retrieve Prior Knowledge from Lessons 3–9

Card 1: L5 Mutation

The Huntington's Locus: What trinucleotide repeat expansion on chromosome 4 causes Huntington's disease, and what is the pathological threshold?

Card 2: L9 Viral Tropism

Vector Selection: Why is Adeno-Associated Virus serotype 5 (AAV5) selected for adult striatal neurons instead of a lentiviral vector?

Card 3: L9 Delivery Routes

Administration: Distinguish between In Vivo, Ex Vivo, and In Situ gene therapy administration routes.

Card 4: L9 BBB Challenge

Surgical Bypass: How does stereotactic Convection-Enhanced Delivery (CED) overcome the Blood-Brain Barrier?

Card 5: L3 Central Dogma

Gene Silencing: At which stage of gene expression (transcription vs translation) does microRNA (miRNA) gene silencing operate?

Card 6: L6 Inheritance

Genetic Risk: Huntington's disease is autosomal dominant. What is the inheritance probability from a heterozygous affected parent?

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Assessment Point 1: Molecular Pathology & RNAi Mechanism

Formative Checkpoint · OCR NEA Task 2 Pass Criteria (P3, P4, P6, P7) & Merit (M2)

Checkpoint 1: Pathology Chain — P3/P4



Pathogenic Cascade: Trace the pathogenic cascade from >36 CAG repeats in HTT exon 1 to polyglutamine misfolding and selective loss of GABAergic medium spiny neurons (MSNs).

Checkpoint 2: Stereotactic CED — P6/P7



Surgical Targeting: Justify why ANT-130 requires bilateral MRI-guided Convection-Enhanced Delivery into the caudate and putamen rather than intravenous injection.

Checkpoint 3: RNAi Silencing — P6/M2



Molecular Cleavage: Explain how Dicer processing and the RISC complex enable engineered miHTT to cleave HTT mRNA and lower toxic mHTT protein.

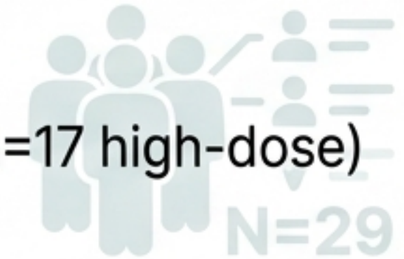
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Learner Activity 1: Auditing NCT04120493 Trial Architecture

Paired Critical Thinking Task (8 Minutes) · Criteria M3 & D2

Task A: Phase I/II Classification

Trial Architecture: Explain why NCT04120493 (29 patients: N=12 low-dose, N=17 high-dose) is a combined Phase I/II trial rather than a pivotal Phase III RCT.



Task B: Sham Surgery Justification

Placebo Control: Discuss why the control arm underwent imitation surgery (superficial cranial bone drilling without dural penetration) to control for placebo effects.



Task C: Unblinding Critique — D2

Data Integrity: Evaluate how unblinding sham control patients after 12 months impacts 36-month comparative data validity.



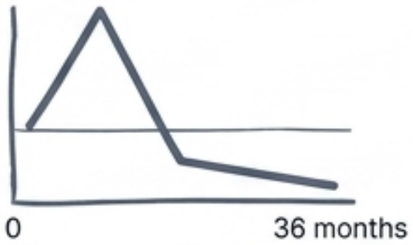
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Learner Activity 2: Interpreting CSF Biomarkers & cUHDRS Data

Data Analysis Workshop (10 Minutes) · Criteria M4, D1 & D2

Challenge 1: The CSF NfL Curve

-8.2%



Biomarker Volatility: Explain why CSF Neurofilament Light (NfL) spikes acutely after surgery before dropping -8.2% below baseline at 36 months.

Challenge 2: Surrogate vs Functional Endpoints

75%
cUHDRS slowing

59%
TMS slowing

Target vs Function: Contrast CSF mHTT lowering (target engagement) against functional preservation (75% cUHDRS slowing, p=0.003; 59% TMS slowing; 60% TFC slowing).

Challenge 3: Statistical vs Clinical Significance



Cohort Comparison: Evaluate why comparing against a propensity score-matched natural history cohort requires Phase III confirmation.

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Assessment Point 2: Evaluating Clinical & Molecular Risks

Diagnostic Risk-Benefit Evaluation · Criteria M3, M4 & D2

Prompt 1: Irreversible Delivery



Permanent Alteration: AAV5-miHTT expression in post-mitotic neurons has 'no undo button.' Evaluate the clinical the clinical risk if delayed neurotoxicity emerges years post-infusion.

Prompt 2: Wild-Type HTT Depletion



Non-Allele Specificity: ANT-130 silences both mutant mHTT and healthy wild-type HTT mRNA.

Why must clinicians monitor long-term wild-type knockdown?

Prompt 3: Immunological Screening



Viral Immunity: Explain why pre-existing AAV5 neutralizing antibodies or neuroinflammation can compromise therapeutic efficacy.

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Learner Activity 3: Genetic Counselling & Bioethics Case Study

Clinical Consultation Roleplay (10 Minutes) · Criteria P8 & M5

Patient Profile



The Scenario: Maya (Age 26) is the asymptomatic daughter of a father newly diagnosed with Huntington's disease (44 CAG repeats). She asks about predictive testing and whether ANT-130 is a complete cure.

Consultation Task 1: Autonomy & Testing — P8

Autonomy & Testing: Advise Maya on her 50% autosomal dominant risk, 100% penetrance, pre-implantation genetic diagnosis (PGD/IVF), and 'The Right Not to Know.'

Consultation Task 2: Managing Expectations — M5

Managing Expectations: Explain clearly why ANT-130 slows progression rather than regenerating lost striatal neurons, and explain sham surgery trial consent.

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Learner Activity 4: NEA Task 2 Distinction Writing Workshop

Coursework Paragraph Drafting & Peer Rubric Audit (12 Minutes) · Criteria D1, D2 & D3

Step 1: Evaluative Synthesis Draft — D2/D3



Drafting: Write a 150-word critical evaluation weighing ANT-130 clinical slowing (75% cUHDRS, -8.2% NfL) against surgical irreversibility and small sample size (N=29).

Step 2: Three-Tier Peer Audit



Highlighting: Swap drafts and highlight evidence meeting Pass (P3–P8: definitions/ mechanisms), Merit (M2–M5: analysis/risks), and Distinction (D1–D3: critical evaluation).

Step 3: Harvard Academic Referencing



Citation Check: Verify in-text citations and reference list entries for uniQure NCT04120493 clinical trial data and peer-reviewed neurology literature.

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